



AUGUST 26 2025

CHE 381 Chemical Engineering Laboratory I

Welcome Back!!



UNIVERSITY OF
ILLINOIS CHICAGO



Today's Agenda

Today we will:

- Go over syllabus & lab schedule
- Discuss safety in the laboratory
 - Safety requirements
 - Chemical and physical hazards in the lab
 - Assign Safety Assignment I
- Assign lab teams and first lab



Chemical Engineering Laboratory I (Unit Operations Lab) Room 128 EIB

Morning: Tuesday 9:00AM – 12:30 PM
Afternoon: Tuesday 1:00PM – 4:30PM

Instructor: Alan Zdunek Office: Room 150 EIB

- Phone: 312-996-4607
- E-mail: zdunek@uic.edu

Teaching Assistant:
Marzia Sultana Email: msult11@uic.edu

Lab Staff: Jan Sagun

Syllabus

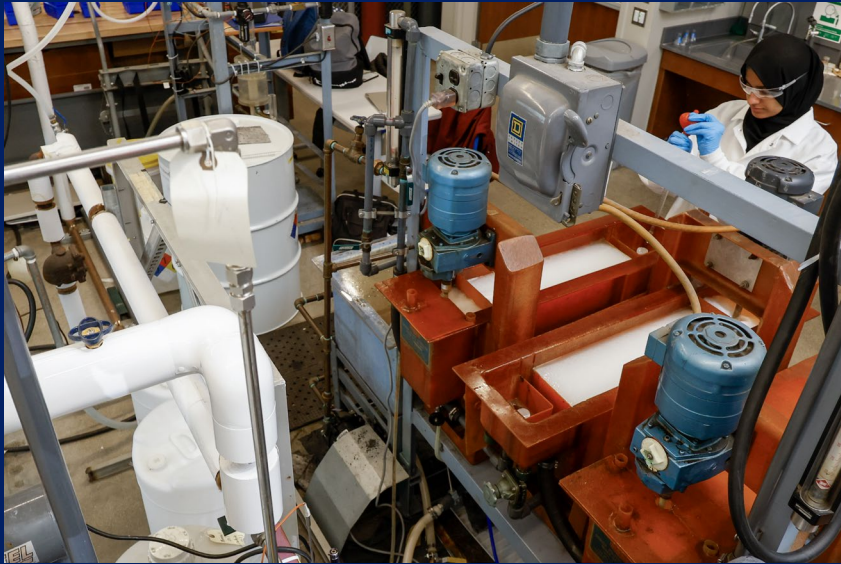


COURSE OBJECTIVES

- Experimental verification of theoretical principles
- Familiarization with unit operations and chemical engineering equipment

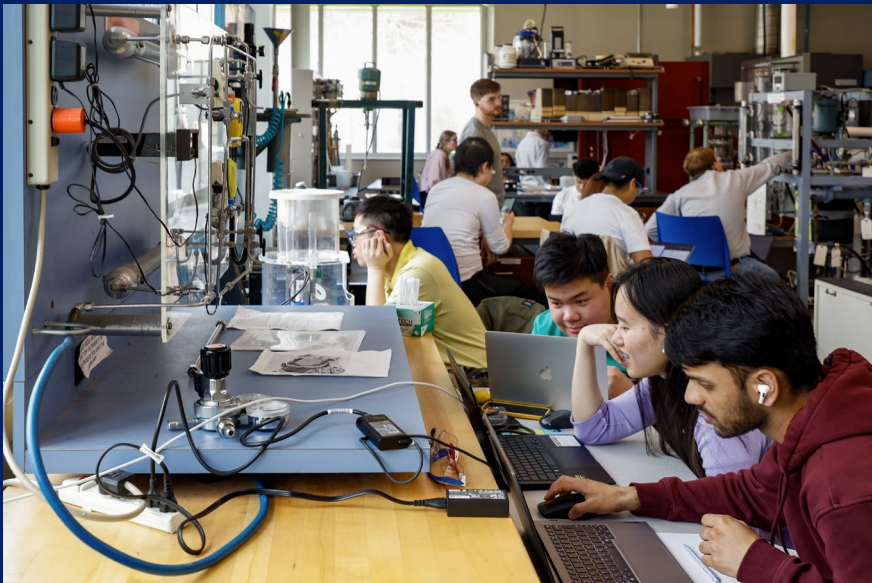
Development of:

- ***Hands-on experience*** and an ***engineering sense***
- ***Teamwork skills***: participation and leadership
- ***Technical communications*** (oral and written)
- ***“Time value”***; project planning, implementation and completion.





Student Learning Objectives

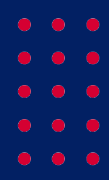


- **Integrate** knowledge and skills from prior chemical engineering courses.
- **Design and conduct** realistic experiments, analyze data, draw concrete conclusions and make relevant recommendations.
- **Solve** complex, open-ended problems creatively, in a team setting.
- **Communicate** effort and findings effectively through the written and oral word.

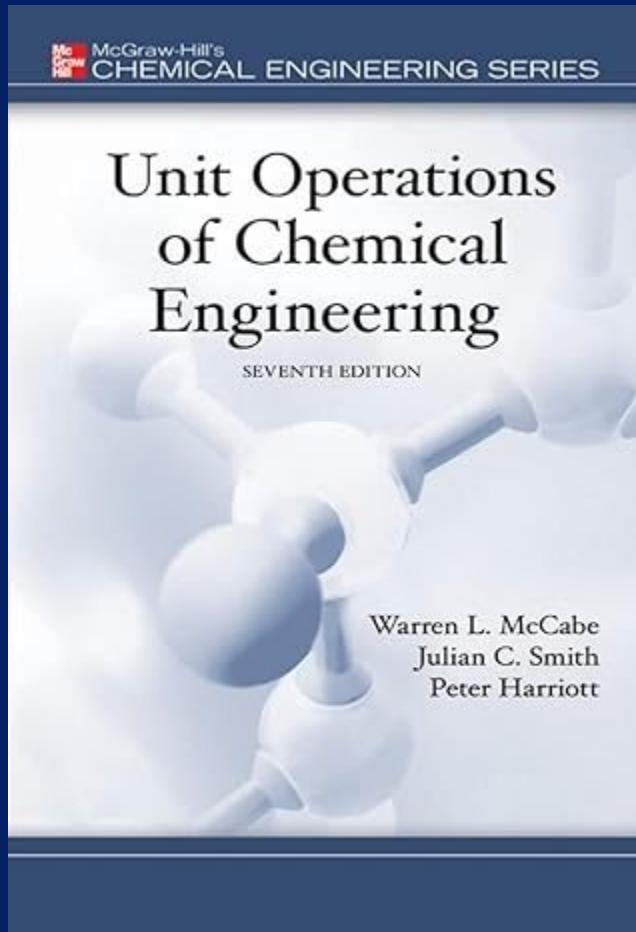
CHE 381 Experiments



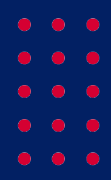
- 1: Non-Newtonian Viscosity
- 2: Evaporation
- 3: Fluid Flow in Pipe Systems
- 4: Humidification
- 5: Membrane Separation
- 6: Distillation



Primary Lab Information



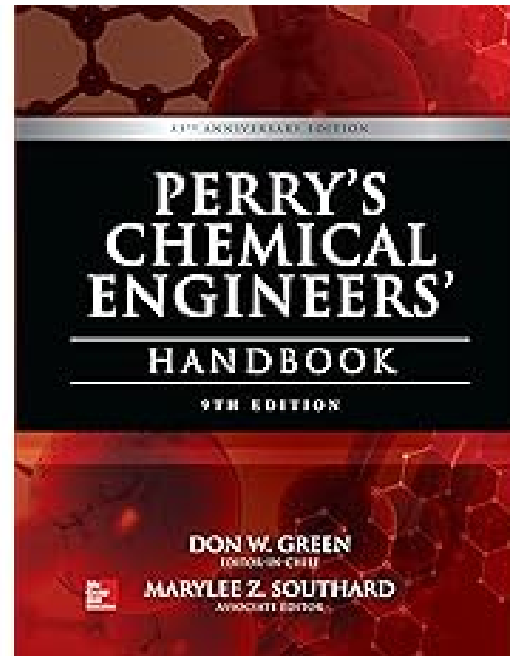
- **Blackboard website** will be used for general class announcements, grades, documents, and any supplementary materials and information.
- **Lab manuals:** Most up-to-date versions available in Blackboard
- **Unit Operations in Chemical Engineering textbook**
 - Great resource for practical information, theory, correlations and report calculations
 - Get a copy and USE IT!!



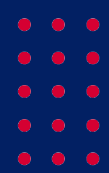
Additional Resources

NOTE: At least 4 references are required for each final lab report (Not including lab manual)

- **Supplemental information** in Blackboard for each experiment
- **Perry's Chemical Engineering Handbook**
 - Another great resource for practical info and correlations.
- **Online resources/UIC library resources**



<https://uic.blackboard.com/>



Lab Schedule



Week 1:	Aug 26	Introductions; lab schedule	
Week 2:	Sept 2	Safety Assignment I Due (BB Assignment)	
Week 2	2	Pre-lab.....	} LAB 1
Week 3	9	Laboratory.....	
Week 4:	16	Pre-lab	} LAB 2
Week 5:	23	Laboratory.....	
Week 6:	30	Pre-lab	} LAB 3
Week 7: Oct	7	Laboratory	
Week 8:	14	Pre-lab	} LAB 4
Week 9:	21	Laboratory	
Week 10:	28	Pre-lab	} LAB 5
Week 11: Nov	4	Laboratory	
Week 12:	11	Pre-lab	} LAB 6
Week 13:	18	Laboratory	
Week 14:	26	Final Presentations.....	EIB 224
Week 15: Dec	5	Safety Assignment II Due No Lab Period	
Week 16: Dec	8-12	FINAL EXAM WEEK	
		No Lab Period	

Course Output



- **Pre-Lab Reports**
- **Final Lab Reports**
- **Oral presentations (every two weeks)**
- **Safety Assignments:**
 - Initial Safety Document
 - Industrial Lab Safety hygiene plan and quiz
 - SACHE/AIChE tutorial

NOTE: Lab Reports are Team Reports

2-Week Experimental Lab Cycle

Two lab periods for each experiment

Prelab Period 1 (Sept 2)

- Familiarization with equipment and experimental objectives
- Assign experimental tasks, results spreadsheet
- Perform initial measurements and calibrations
- Prepare Pre-Lab Report with Safety Analysis

Prelab Report DUE: Next day after prelab period, upload into Blackboard assignments (September 3rd for Exp #1)

Final Lab Period 1 (Sept 9)

Perform & complete experimental work
Collect & analyze, create graphs, figures and tables, error analysis.
Begin Final Lab Report

In-Between Lab Periods

- Analyze results and write up final lab report
- Put together oral presentation

Final lab Report DUE: Next the Monday upload into Blackboard assignments (September 15th for Experiment #1)

Prelab Period 2 (Sept 16)

Oral presentation to next lab team or TA/Instructor
Begin Prelab for Experiment #2

NOTE: USE WORD DOCUMENT FOR LAB REPORTS!



1st Lab Schedule Summary



1st pre-lab period will start next week

Tuesday September 2nd

1st Pre-Lab Plan due:

Wednesday September 3rd

End of Day: 11:59pm

1st Final Lab period: September 9th

1st final lab report due Monday Sept 15th

End of Day: 11:59pm

1st Lab Presentation due Tuesday 9/16

NOTE: USE WORD DOCUMENT FOR LAB REPORTS!



Attendance



- Attendance is required for all lab periods.
 - On-time: ≤ 15 minutes
 - Late: ≥ 15 minutes to ≤ 50 minutes
 - Absent: > 50 minutes
- One excused absence will be allowed and considered on an individual basis, but **you must personally contact instructor at least 24 hours before the lab period, otherwise it will be an unexcused absence. Email is best.**
- **Two unexcused absences or 4 late marks will result in a zero on the individual assessment grade.**
- Attendance will be part of individual assessment grade.

Pre-Lab Report



**NOTE: USE WORD
DOCUMENT FOR LAB
REPORTS!**

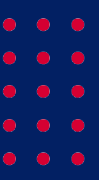
- Briefly state the objective of the experiment, summarize the apparatus and materials & supplies that will be used, explain the experimental procedure to be followed and conduct a safety analysis of the equipment.

(Length: 4-5 pages)

Lab Prep Report Sections:

- Title page, Title, Lab #, Team #, Team member names
- **Experimental: Objective, Apparatus, Materials/Supplies, Experimental Procedure**
- **Safety Analysis Sheet**

See PreLabPlanTemplate in Blackboard for more details



Final Lab Report

Executive Summary (industry format)

Introduction

Theory

Results

Discussion

References

Appendices

Data Tabulation (Raw data)

Error Analysis

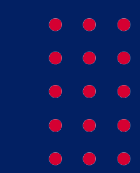
Sample Calculations

Individual Team Contributions

- Tells what you did, why you did it, gives main analysis and results, states what you learned, the significance of your results and your conclusions.
- **Length 10-15 pages** (not including Appendices (Raw Data, Error Analysis, Calculations, Individual Contributions))
- Key to writing an excellent report lies in presenting concisely your response to the objectives of the project and justifying that response with appropriate documentation.

**NOTE: USE WORD
DOCUMENT FOR LAB
REPORTS!**

**See FinalLabReportTemplate in Blackboard for
more details**



Oral Presentations

Introduction

Theory

Apparatus

Materials/Supplies

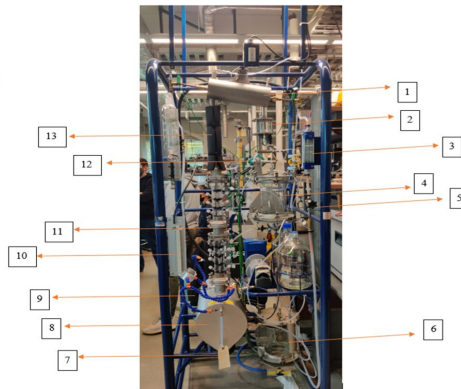
Experimental Procedure

Experimental Challenges

Safety

Apparatus

Number	Equipment
1	Condenser
2	Decanter
3	Cooling water control
4	Top product receiver
5	Cooling water inlet valve
6	Bottom product receiver
7	Bottoms product sample valve
8	Reboiler
9	Heater
10	Manometer
11	Distillation Column
12	Reflux regulator valve
13	Insulation



- After completing an experiment, the team will make a 10-minute powerpoint presentation to the next team or the instructor/TA (next pre-lab period).
- Brief but concise presentation describing:
 - Objective
 - Experimental procedures
 - Experimental challenges, safety
 - Lessons learned
- Use MS presentation software (see template in BB)
- Upload to Blackboard in Assignments: Due at end of Prelab period.

See Presentation Template in Blackboard for more details

Laboratory Safety



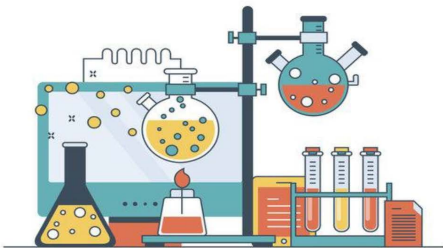
Laboratory Safety Plan

Principal Investigator _____

Chemical Hygiene Officer _____

Lab Room Number _____

2025



Informational Services	Time	(312) Area Code
Emergency- Police / Fire	24 hours/Day	312-355-5555
Emergency – Chemical/Biological	24 hours/Day	996-7233
Environmental Health and Safety Office	Normal Business Hours	996-7411
	After Business Hours	996-7233 (6-SAFE)
	Email	labsafety@uic.edu
UIC (Hospital) Emergency Service	24 Hours/Day	996-7297
UIC Health Service	Normal Business Hours	996-7420
Illinois Poison Center	24 Hours/Day	1-800-942-5969 1-800-222-1222
Eye and Ear Infirmary	Normal Business Hours	996-6562
Facilities Emergency Services	24 Hours/Day	996-7511
Your Department Office Telephone	FILL IN →	
Your Building Number	FILL IN →	

Heather Jackson, Director of the Environmental Health & Safety Office (EHSO)	413-9706
Kevin Cisner, Associate Director, Research, Chemical and Radiation Safety	413-3387
Changhwa Hwang, Associate Director for Biological Safety	996-6873
Scott Dubick, Assistant Director of Industrial Hygienist	996-1178
Mark Gaunky, Assistant Director of Fire and Life Safety	996-7116
Carlos Lopez, Research Safety Manager	413-5059
Tom Johnson, Hazardous Waste Compliance Manager	413-9086
Jose Gonzalez, Environmental Engineer	413-7260

Lab Safety Details



- Compressed air supply (house air):
 - 120 psig supply pressure
 - Regulators to reduce pressure
- Compressed air gas cylinder
 - Very high pressure: 2000 psig
 - Two-stage regulator
 - Ask TA or Instructor for guidance before using.
- Chemical usage (Ethanol, NaOH, Carbopol powder)
 - Read MSDS sheets to familiarize yourself with hazards
 - Wear proper safety attire when handling (chemical-resistant gloves → different than blue nitrile lab gloves)



Lab Safety Details



Steam
generator



Fume Hood

- High-temperature steam (humidification, evaporation experiments)
 - Pressure: 70 psig regulated down to 10-15 psig
 - Temperature: 113°C
- Know where the safety shower, eyewash and fire extinguishers, first aid kit, MSDS are located
- Use fume hoods for chemical handling
 - Ethanol for Distillation experiment



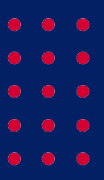
Lab Safety Guidelines

Initial Safety Document: Read, sign and return at first lab period

1. You must wear standard safety glasses in the lab at all times. You must wear chemical workers' goggles for any operation with hazardous chemicals.
2. Adequate leg protection is required (no shorts).
3. Full foot protection is required (no open-toed shoes; try boots if possible).
4. Use disposable nitrile gloves for hand protection when handling any chemical.
5. If you wear contact lenses, you must wear chemical workers' goggles in the lab at all times.
6. In the event of a fire or other accident, contact your TA immediately. Please review the location of the laboratory fire extinguishers. Know how to call emergency services.
7. Cuts or burns must be treated immediately at the student health center. Transportation will be arranged for you.
8. Neither food nor drink is allowed in the laboratory.
9. Please use rubber suction to fill pipettes.
10. Secure long hair, ties(?), or loose jewelry. Long unrestrained items can become entangled in moving equipment. This could damage equipment.
11. Unauthorized experiments are not allowed (Do that stuff at home). Never work in the laboratory alone. No experimental work outside of lab times is allowed unless authorized by their instructor or TA.
12. Be sure every electrical device is plugged in and well grounded before you use it. When in doubt, please ask!

- Adequate leg protection (no shorts) is required
- Eating and drinking are not allowed in laboratory!!
- Safety glasses must always be worn in the lab at all times.
- You will be given your own personal pair to be used all year.
- **Safety glass and food/drink compliance will be monitored by the instructor and TA during the lab period.** Non-compliance will be recorded.
 - 3 instances of non-compliance will result in a zero on the Individual Assessment grade.





Safety

Assignment I:

Industrial

Chemical

Hygiene Plan

- **Chemical Hygiene Plan (CHP) and Safety Rules from a corporate R&D Center**
- **CHP's are a required OSHA safety document at R&D facilities.**
- **Blackboard Safety Assignment I**



Safety

Assignment I

Due next

Tuesday: 9/2/25

- READING ASSIGNMENT: You will be responsible for material in the following Chapters: Chapter 1, 3.2, 4, 5, 6, 7, 8. and the Definitions at the End. *SKIP Chapter 2, 3.1 and 3.3 and the Appendices and Attachments (not included).*
- SAFETY QUIZ: CHE 381 Safety Quiz Fall 2025.doc can be found in the Blackboard assignment. Be brief but concise in your answers. Open book/open notes and team discussion is allowed but you must fill out and submit the quiz yourself.





Safety

Assignment II: SACHE Safety Tutorial

AICHE (American Institute of Chemical Engineers) has created an online academy with safety tutorials covering process safety, risk assessment and chemical reactivity hazards.



- ASSIGNMENT: Complete tutorial “**Chemical Reactivity Hazards**” ELA962 from the website.
- **NOTE: You must be a student member to get the tutorial for FREE.**
- GRADING: You must upload the Certificate issued that you have completed the tutorial and answer the questions for full credit.
- **DUE: Friday December 5th at 11:59 pm**



Grade Determination

Pre-Lab Plan	25%
Final Lab Report	40%
Oral Presentations	5%
Safety quiz/ SACHE tutorial	10%
<u>Individual Performance Assessment</u>	<u>20%</u>
	100%



Academic Integrity & Plagiarism

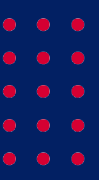
As a student and member of the UIC community, you are expected to adhere to the Community Standards of integrity, accountability, and respect in all your academic endeavors.

Plagiarism: “the unacknowledged inclusion of someone else’s words, structure, ideas, or data.”

- Learn when and how to cite sources correctly. A helpful website is “Plagiarism.org”.
- SafeAssign in Blackboard

Unacceptable behavior:

- Cheating
- Unauthorized collaboration
- Fabrication or falsification
- Plagiarism
- Using unauthorized study aids
- Coercion regarding grading or evaluation of coursework



Academic Integrity & Plagiarism

- The behaviors listed will not be tolerated and will result at a minimum in a zero for the report, presentation, or assignment.
- Other possible actions: An “F” on the final grade and investigation and disciplinary action by the Office of the Dean of Students
- In egregious cases, expulsion from the university.

Lab Team Setup



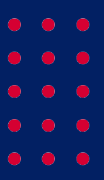


Tuesday AM

**Lab Teams:
Four teams of
4 members**

Lab 1 Experiments

- **Team 1: Non-Newtonian Viscosity**
- **Team 2: Evaporation**
- **Team 3: Fluid Flow in Pipes**
- **Team 4: Humidification**



Lab 1 Experiments

Tuesday PM

Lab Teams:

- 2 Teams of 3 members
- 1 Team of 4 members

- Team 1: Non-Newtonian Viscosity
- Team 2: Evaporation
- Team 2: Humidification